Human Health and Disease

Introduction and Common Diseases in Human

1. The term 'Health' is defined in many ways. The most accurate definition of the health would be
1) Health is the state of body and mind in a balanced condition
2) Health is the reflection of a smiling face
3) Health is a state of complete physical, mental, and social well-being
4) Health is the symbol of economic prosperity.

2. The organisms which cause diseases in plants and animals are called
1) Pathogens
2) Vectors
3) Insects
4) Worms

3. Which of the following set of diseases is caused by bacteria?
1) Typhoid and smallpox
2) Typhoid and mumps
3) Cholera and tetanus
4) Herpes and smallpox

4. A patient suffering with malaria experiences recurring chills and fever at the time when
1) The microgametocytes and macrogametocytes are destroyed by WBCs
2) The trophozoites reach maximum growth and give out certain toxins
3) The sporozoites released from RBCs are being rapidly killed and broken down in spleen
4) The parasite after its rapid multiplication inside RBCs ruptures them, releasing the stage that enters fresh RBCs.

5. A doctor identifies symptoms of nasal congestion, headache, sore throat, hoarseness, cough in a patient. According to his diagnosis, the patient is infected with
1) Salmonella
2) Rhinovirus
3) Yersinia pestis
4) Plasmodium

6. Match the columns and choose the correct option.

Column I		Column II	
a	Haemophilus influenzae	p	Malignant malaria
b	Entamoeba histolytica	q	Elephantiasis
c	Plasmodium falciparum	r	Pneumonia
d	Wuchereria bancrofti	s	Typhoid
e	Salmonella typhi	t	Amoebiasis

1) a-p, b-t, c-r, d-q, e-s
2) a-r, b-t, c-p, d-q, e-s
3) a-t, b-p, c-r, d-s, e-q
4) a-p, b—r, c—t, dq, e-s

7. Which of the following is incorrect w.r.t. amoebiasis?
1) It is caused by Entamoeba histolytica which is a protozoan parasite of small intestine of humans.
2) Housefly acts as the mechanical carrier and serves to transmit the parasite from faeces of infected persons to food and food products.

3) Symptoms of amoebiasis include constipation, abdominal pain and cramps, stools with excess mucous and blood clots.
4) It is also called amoebic dysentery.

8. Which of the following is/are incorrect w.r.1. typhoid?
(i) The pathogen enters the human body through contaminated food and water.
(ii) Sustained high fever, weakness, stomach pain, constipation, headache and loss of appetite are common symptoms.
(iii) Widal test helps in diagnosis of this disease.
(iv) The disease is caused by Streptococcus pneumonia.
(v) Intestinal perforation and death can occur in severe cases.
1) Only (i)
2) Only (iv)
3) (ii) and (iii)
4) (ii) and (iv)

9. Among the non-infectious diseases, the major cause of death is
1) Diabetes
2) Cancer
3) Mumps
4) Hypertension

10. Swelling of body parts like limbs occurs in elephantiasis due to
1) Deficiency of proteins causing oedema
2) Renal failure
3) Heart failure
4) Blockage of lymphatic circulation by filarial worm

11. Close contact with the infected persons or their belongings should be avoided in
1) Elephantiasis
2) Pneumonia and common cold
3) Cancer
4) Malaria

12. Mark the wrong statement.
1) Ascariasis, Taeniasis and Enterobiasis are helminthic diseases.
2) Measles, Mumps and Rubella are viral diseases.
3) Malaria, Amoebiasis and Filariasis are protozoan diseases.
4) Cholera, Diphtheria and Tetanus are bacterial diseases.

13. Which of the following is an incorrect statement?
1) Typhoid is caused by Salmonella typhi.
2) Trophozoite is the infective stage of malarial parasite for humans.
3) Filarial worm is transmitted by Culex mosquito.
4) Infection of Ascaris occurs by consuming food or water carrying its eggs.

14. Given below is a list of some diseases:

Measles, Mumps, Rubella, Tetanus, Hypertension, Rickets, Plague, Dengue fever, Cancer, Pneumonia, Scurvy, Haemophilia, Pertussis, Leprosy, Typhoid

How many of these are infectious diseases caused by bacteria?
1) 6
2) 7
3) 8
4) 9

15. Which of the following are correct statements?

(i) Ringworm infection is characterized by dry, scaly lesion on various parts of the body such as skin, nails and scalp.
(ii) Plasmodium initially infects RBCs and then infects liver cells.
(iii) Amoebiasis is caused by a protozoan pathogen called Amoeba proteus.
(iv) Wuchereria is an intestinal parasite causing internal bleeding, muscular pain, fever, anaemia and blockage of the intestinal passage.
(v) Ringworms are generally acquired from soil or by using towels, clothes or even the comb of infected individuals.

1) (i) and (ii)
2) (ii) and (iv)
3) (i) and (v)
4) (iv) and (v)

16. Identify the odd one w.r.t. helminthic diseases.
1) Taeniasis
2) Ascariasis
3) Enterobiasis
4) Amoebiasis

17. Pneumonia is caused by Streptococcus pneumoniae and
1) Rhinovirus
2) Haemophilus influenzae
3) Salmonella typhi
4) Clostridium tetani

18. Read the following statements describing the life cycle of Plasmodium. Which of the following is correct?
1) Fertilization and development take place in liver of humans.
2) Female mosquito takes up sporozoites with blood meal.
3) The gametocytes develop in intestine of humans.
4) The parasite reproduces asexually in RBCs and causes haemolysis and fever.

19. Read the following statements and choose the incorrect option.
1) The eggs of the Ascaris are excreted along with the faeces of infected persons which contaminate soil, water, plants, etc.
2) Mosquito act as mechanical carriers of amoebiasis and serve to transmit the parasite from faeces of infected person to food and food products, thereby contaminating them.
3) The pathogens can enter our body by various means, multiply and interfere with normal vital activities, resulting in morphological and functional damage.
4) Salmonella typhi generally enter the small intestine through food and water contaminated with them and migrate to other organs through blood.

20. Mark the incorrect option w.r.t communicable diseases.
1) Mumps, Rubella
2) AIDS, Chicken pox
3) Leprosy, Typhoid
4) Haemophilia, Asthma

Immunity

21. Find the incorrect statements.
1) Helper T-cells stimulate the B-cells to divide and produce antibodies.
2) T-cells attack transplanted tissue cells.

3) Vaccines work on the principle of memory.
4) Among carbohydrates, fats, proteins and nucleic acids, only proteins can act as antigens and elicit an immune response.

22. Which of the following is an incorrect statement w.r.t. secondary response?
1) It is also called anamnestic response.
2) It is highly intensified.
3) It is based on the fact that there is memory of previous infection.
4) It fades quickly and is of low intensity.

23. The substance produced by a cell in viral infection that can protect other cells from further infection is
1) Serotonin
2) Colostrum
3) Interferon
4) Histamine

24. Match the column I with column II, and choose the correct option.

Column I		Column II	
a	Cellular barrier	1	Natural killer
b	Physiological barrier	2	Interferons
c	Cytokine barrier	3	Skin
d	Physical barrier	4	Tears from eyes

1) a-2, b-3, c-4, d-1
2) a-1, b-4, c-2, d-3
3) a-3, b-2, c-4, d—1
4) a-1, b-3, c-2, d- 4

25. Read the following statements describing innate immunity:
(i) They are non-specific type of defense.
(ii) They do not have memory component.
(iii) They develop only after exposure to pathogen.
(iv) These are defences that are present from birth.
Which of the above statement(s) is/are incorrect?
1) (ii) only
2) (ii) and (iii)
3) (iii) only
4) (iii) and (iv)

26. Transplantation of tissues/organs to save certain patients often fails due to rejection of such tissues/organs by the patient. Which type of immune response is responsible for such rejections?
1) Auto-immune response
2) Humoral immune response
3) Physiological immune response
4) Cell-mediated immune response

27. Read the following statements.
Statement A: If due to some reason, B-and T-lymphocytes are damaged, the body will not be able to produce antibody against a pathogen.
Statement B: Injection of dead/inactivated pathogens causes passive immunity.
Choose the correct option:
1) Statement B is correct and statement A is wrong.
2) Both the statements A and B are correct.
3) Statement A is correct and statement B is wrong.

4) Both the statements A and B are wrong.

28. All of the following are physiological barriers that prevent microbial growth, except

1) Saliva in mouth
2) Tears from eyes
3) Acid in stomach
4) Mucus coating of the epithelium

29. Read the following statements and identify true (T) and false (F) among them:

(i) T-lymphocytes mediate humoral immune response.

(ii) Mucus coating of the epithelium lining the respiratory and urogenital tract form physical barrier of innate immunity.

(iii) Bacteria-infected cells secrete proteins called interferons which protect non-infected cells from further infection.

(iv) Preformed antibodies are given to victims of snakebite which triggers active immunity in them.

	(i)	**(ii)**	**(iii)**	**(iv)**
1)	F	F	F	T
2)	T	F	F	F
3)	F	T	F	F
4)	T	T	F	F

30. Read the following statements:

(i) PMNL and monocytes are included in cellular barrier of innate immunity.

(ii) The principle of vaccination is based on the memory of immune system.

(iii) Skin on our body belongs to the same category of the barriers of innate immunity to which acid in stomach also belongs.

(iv) Vaccination is an example of passive immunity.

Which of the above statements is/are correct?

1) (i) and (ii)
2) (ii) and (iii)
3) (ii) and (iv)
4) (i) and (iii)

31. Which of the following is not an example of artificially acquired passive immunity?

1) Antivenom given to person who had snakebite.
2) IgG received by foetus across placenta.
3) Anti-tetanus serum given to an injured patient.
4) Anti-rabies serum given to a patient after dog bite.

32. In which of the following organs, the lymphocytes interact with antigens, undergo proliferation and get differentiated into effector cells?

1) Primary lymphoid organ
2) Bone marrow
3) Thymus
4) Secondary lymphoid organ

33. Passive immunity is defined as immunity

1) Developed through vaccination
2) Obtained from sera of other animals enriched in antibodies
3) Obtained by foetus from its mother through placenta
4) Both (2) and (3)

34. Which of the following is not a lymphoid tissue?

1) Spleen

2) Tonsils
3) Pancreas
4) Thymus

AIDS

35. Which of the following individuals are at high risk of getting infected with HIV?
1) Intravenous drug abusers
2) People with multiple sexual partners
3) People who require repeated blood transfusion
4) All of the above

36. World AIDS day, observed on ____ every year, is dedicated to raising awareness of the AIDS pandemic caused by the spread of HIV infection and mourning those who have died of the disease.
1) 11th July
2) 1st December
3) 6th March
4) 21th August

37. Transmission of HIV-infection generally occurs by
I. Sexual contact with infected person
II. Transfusion of contaminated blood and blood products
III. Sharing infected needles as in the case of intravenous drug abusers
IV. From infected mother to her child through placenta
1) All of the above
2) I, II and IV
3) Only I
4) I, III and IV

38. Read the assertion and reason statements carefully and select the correct option.
Assertion: Infection with HIV or having AIDS is something that should not be hidden.
Reason: The infection may spread to many more people due to lack of awareness.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false

39. Which of the following is incorrect?
1) HIV is not spread by mere touch or physical contact, and spreads only through body fluids.
2) A time lag between HIV infection and appearance of AIDS symptoms may vary from few months to many years (usually 5-10 years).
3) After getting into the body of the person, the virus enters into macrophages where DNA genome of the virus replicates to form RNA with the help of reverse transcriptase.
4) A widely used diagnostic test for AIDS is ELISA.

40. A person with AIDS is prone to infection by
1) Toxoplasma
2) Fungi
3) Mycobacterium
4) All of the above

41. After getting into the body of the person, the HIV enters the macrophages where RNA genome of the virus replicates to form viral DNA with the help of the enzyme

1) Reverse transcriptase
2) DNA polymerase
3) Topoisomerase
4) DNA gyrase

42. Which of the following cells are called HIV factory?
1) Neutrophils
2) Macrophages
3) Basophils
4) T-lymphocytes

43. Anti-retroviral therapy is given to HIV positive patients with drugs like azidothymidine. These drugs
1) Can completely cure the patient
2) Are totally ineffective
3) Can prolong life and improve quality of life but cannot prevent death
4) Can eliminate HIV from body if treatment is started within 2 weeks of infection

44. In our country, NACO and other non-governmental organizations (NGOs) are doing a lot to educate people about AIDS. NACO stands for
1) National Addiction Control Organization
2) National AIDS Control Organization
3) National Awareness and Control Organization
4) National AIDS and Cancer Organization

45. A widely used diagnostic test for AIDS is
1) Widal test
2) Serum and urine analysis
3) ELISA
4) Biopsy

Cancer

46. Which of the following is an incorrect statement w.r.t. cancer cells?
1) Cancer cells detach from tumours and spread to other parts of the body by a property called metastasis.
2) Cancer cells rapidly divide and starve the normal cells by competing for vital nutrients.
3) Benign tumours grow very rapidly and invade into the surrounding normal tissues, while malignant tumor cells grow slowly and remain confined to the tissue of its origin.
4) In cancer cell, there is a breakdown of the regulatory mechanism of cell growth and differentiation.

47. Which of the following non-ionizing radiations cause DNA damage leading to neoplastic transformation?
1) X-rays
2) Gamma rays
3) Alpha particles
4) UV rays

48. The genes causing cancer are
1) Structural genes
2) Expressor genes
3) Oncogenes
4) Regulatory genes

49. Tumours are caused due to
1) Loss of contact inhibition
2) Uncontrolled proliferation of cells
3) Failure in regulation of cell division
4) All of these

50. Which of the following statements are correct regarding cancer?
(i) Mutations in tumour suppressor genes are linked with cancer.
(ii) Cancer cells show the property of contact inhibition.
(iii) Radiotherapy involves irradiation of cancer cells to destroy their genetic material.
(iv) Alpha interferons are used for treatment of cancer.
(v) Non-ionizing radiations like X-rays and gamma rays cause DNA damage leading to neoplastic transformation.
1) (i), (ii) and (iii)
2) (i), (iii) and (iv)
3) (ii), (iii) and (v)
4) (iii), (iv) and (v)

51. To generate a three-dimensional image of the internals of an object, computed tomography uses
1) UV rays
2) X-rays
3) Gamma rays
4) Infrared rays

52. In malignant tumors, the cells proliferate, grow rapidly, and move to other parts of the body to form new tumors. This stage of disease is called
1) Metagenesis
2) Metastasis
3) Teratogenesis
4) Mitosis

53. Read the given statements and choose the correct option
Statement I: Metastatic cancers are more serious.
Statement II: These spread from one organ to other body organs and there is increased interference with metabolic functioning.
1) Both statement I and statement II are true
2) Both statement I and statement II are false
3) Statement I is true, and statement II is false
4) Statement I is false, and statement II is true

54. Which of the genes when activated in a normal human body can cause cancer?
1) Proto-oncogenes
2) c-oncogenes
3) Cellular oncogenes
4) All of these

55. To accurately detect pathological and physiological changes in the living tissues, MRI uses
1) Strong magnetic fields and non-ionising radiations
2) Strong magnetic fields and ionising radiations
3) Weak magnetic fields and non-ionising radiations
4) Weak magnetic fields and ionising radiations

56. Cancer cells are more easily damaged by radiation than normal cells because they are
1) Non-dividing
2) Different in structure

3) Undergoing rapid division
4) Mutated

57. Read the following statements regarding cancer. Which of the following are correct?

(i) The technique involving excision of a part of tumour followed by its microscopic examination is called ultrasonography.
(ii) The side effects of chemotherapy are hair loss and anaemia.
(iii) Carcinoma refers to tumour of skin or epithelial lining.
(iv) The scientific study of cancers is called toxicology.

1) (i) and (ii)
2) (ii) and (iii)
3) (iii) and (iv)
4) (i) and (iv)

Drugs and Alcohol Abuse

58. Which of the following is incorrect w.r.t. drug heroin?

1) It is commonly called smack.
2) It is chemically dimethylmorphine.
3) It is a white, odourless, bitter crystalline compound.
4) It is a depressant and slows down body functions.

59. Which of the following is an incorrect statement w.r.t. opioids?

1) Opioids bind to specific receptors present in our central nervous system and gastrointestinal tract.
2) They are obtained from Papaver somniferum.
3) Heroin is a natural alkaloid obtained from leaves of poppy plant.
4) Opioids are generally taken by snorting or injection.

60. Mark the wrong statement:

1) Marijuana is obtained from Cannabis plant.
2) Caffeine is obtained from Atropa belladonna.
3) Cocaine has a potent stimulating action on CNS, producing sense of euphoria and increased energy.
4) These days cannabinoids are abused by some sportspersons.

61. ´Smack' is a drug obtained from the

1) Latex of Papaver somniferum
2) Leaves of Cannabis sativa
3) Flowers of Datura
4) Fruits of Erythroxylum coca

62. Read the following statements and select the correct option.

Statement I: Tobacco has been used by human beings for more than four centuries.
Statement II: Smoking is associated with increased incidences of cancers of lung, urinary bladder and throat, bronchitis, emphysema, coronary heart disease, etc.

1) Both statement I and statement II are true
2) Both statement I and statement II are false
3) Statement I is true, and statement II is false
4) Statement I is false, and statement II is true

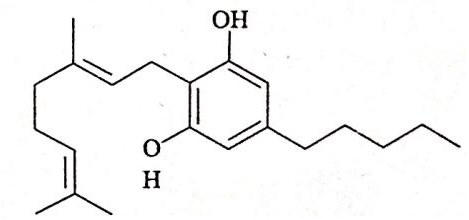
63. Which of the following is an incorrect statement?

1) Tobacco contains a number of chemical substances including nicotine, an alkaloid.

2) Nicotine stimulates adrenal gland to release adrenaline and noradrenaline, which increases blood pressure and heart rate.
3) Tobacco chewing is associated with increased risk of cancer of the oral cavity.
4) Smoking reduces carbon monoxide content in blood and increases the concentration of haem-bound oxygen.

64. Read the following statements and select the correct option.
Statement A: Morphine is sedative and painkiller.
Statement B: Barbiturates cause mental retardation.
1) Statement B is correct and statement A is wrong.
2) Both the statements A and B are correct.
3) Statement A is correct and statement B is wrong.
4) Both the statements A and B are wrong.

65. Identify the given structure.



1) Morphine
2) Cannabinoid molecule
3) Aspirin
4) Cholesterol

66. Which of the following drugs are obtained from the same source?
1) Morphine and Charas
2) Charas and Marijuana
3) Charas and Cocaine
4) Morphine and Marijuana

67. Bhang, charas, ganja, hashish and marijuana are drugs obtained from
1) Poppy plant
2) Hemp plant
3) Datura
4) Ergot fungus

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	1	3	4	2	2	1	2	2	4	2	3	2	1	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	2	4	2	4	4	4	3	2	3	4	3	4	3	1
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	4	4	3	4	2	1	1	3	4	1	2	3	2	3
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
3	4	3	4	2	2	2	1	4	1	3	2	2	3	2
61	**62**	**63**	**64**	**65**	**66**	**67**								
1	1	4	3	2	2	2								